

Paolo Bruschi

professor

基本信息

姓名: Paolo Bruschi

所属院校: 比萨大学

地区: Italy

年龄范围: 50岁以上

学科领域

微电子科学与工程

集成电路设计与集成系统

测控技术与仪器

智能感知工程

研究领域

传感器原理与测量技术

模拟与数模混合电路

混合信号集成技术

低功耗电路与能量管理

模拟与射频IC设计

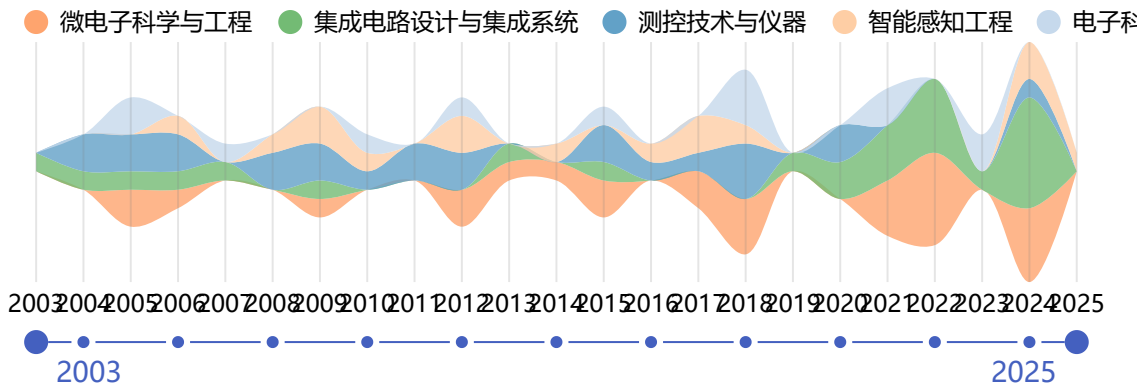
信号调理与数据采集

微纳传感器与MEMS

个人简介

Paolo Bruschi is a Full Professor in the Dept. of Information Engineering at the University of Pisa. His research interests include Analog Integrated Circuit Design, Sensors, Sensor systems, and MEMS. He has numerous publications, such as "A low-cost sensing system for cooperative air quality monitoring in urban areas" in Sensors (2015, cited 118 times), "Temperature-dependent Monte Carlo simulations of thin metal film growth and percolation" in Physical Review B (1997, cited 106 times), and many others across different journals and conferences.

近年发表的论文



An ASIC-based system-in-package MEMS gas sensor with impedance spectroscopy readout and AI-enabled identification capabilities

SENSORS AND ACTUATORS B-CHEMICAL

Zampolli, S.; Elmi, I.; Bruschi, P.; Ria, A.; Magliocca, F.; Vitelli, M.; Piotto, M.

被引用次数: 0 2025

New One Data Wire Serial Bus for Distributed Sensor Arrays

IEEE SENSORS JOURNAL

Balocchi, Leonardo; Vitelli, Michele; Contardi, Simone; Nannipieri, Iacopo; Bruschi, Paolo; Bonafoni, Stefania; Roselli, Luca

被引用次数: 0 2024

Parallel Slew-Rate Enhancer With Current-Recycling Core for Switched-Capacitors Circuits

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Gagliardi, Francesco; Catania, Alessandro; Piotto, Massimo; Bruschi, Paolo; Dei, Michele

被引用次数: 0 2024

A miniaturized ECG system based on a versatile single chip sensor interface

2024 9TH INTERNATIONAL CONFERENCE ON SMART AND SUSTAINABLE TECHNOLOGIES, SPLITECH 2024

Ria, Andrea; Contardi, Simone; Piotto, Massimo; Bruschi, Paolo

被引用次数: 0 2024

Static-Linearity Enhancement Techniques for Digital-to-Analog Converters Exploiting Optimal Arrangements of Unit Elements

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Gagliardi, Francesco; Scintu, Danilo; Piotto, Massimo; Bruschi, Paolo; Dei, Michele

被引用次数: 0 2024

A Fully Integrated 1 kHz-1 MHz Sinusoidal Waveform Generator based on Second Order Analog Interpolation

2024 19TH CONFERENCE ON PH.D RESEARCH IN MICROELECTRONICS AND ELECTRONICS, PRIME 2024

Scognamiglio, Margherita; Gagliardi, Francesco; Contardi, Simone; Nannipieri, Iacopo; Bruschi, Paolo

被引用次数: 1 2024

A Compact Fault Tolerant 2-D Anemometer

IEEE SENSORS JOURNAL

Ria, Andrea; Bruschi, Paolo; Piotto, Massimo

被引用次数: 0

2024

Ultralow-Power Inverter-Based Delta-Sigma Modulator for Wearable Applications

IEEE ACCESS

Catania, Alessandro; Gagliardi, Francesco; Piotto, Massimo; Bruschi, Paolo; Dei, Michele

被引用次数: 1

2024

A 114 ppm/°C-TC 0.78%-(σ/μ) Current Reference With Minimum-Current-Search Calibration

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Gagliardi, Francesco; Ria, Andrea; Piotto, Massimo; Bruschi, Paolo

被引用次数: 2

2024

Online High-Resolution EIS of Lithium-Ion Batteries by Means of Compact and Low Power ASIC

BATTERIES-BASEL

Ria, Andrea; Manfredini, Giuseppe; Gagliardi, Francesco; Vitelli, Michele; Bruschi, Paolo; Piotto, Massimo

被引用次数: 1

2023

A 1-GS/s 6-8-b Cryo-CMOS SAR ADC for Quantum Computing

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Kiene, Gerd; Overwater, Ramon W. J.; Catania, Alessandro; Sreenivasulu, Aishwarya Gunaputi; Bruschi, Paolo; Charbon, Edoardo; Babaie, Masoud; Sebastiano, Fabio

被引用次数: 8

2023

Performance Modelling of Optimal Combination Algorithms Applied to Arbitrary Data Converter Architectures

2023 IEEE NORDIC CIRCUITS AND SYSTEMS CONFERENCE, NORCAS

Gagliardi, Francesco; Scintu, Danilo; Piotto, Massimo; Bruschi, Paolo; Dei, Michele

被引用次数: 1

2023

Low-Cost Sweating-Rate Sensor for Dehydration Monitoring in Sports

2023 IEEE SENSORS

Ria, Andrea; Piotto, Massimo; Munoz-Berbel, Xavier; Bruschi, Paolo; Dei, Michele

被引用次数: 2

2023

A Codesigned Integrated Photonic Electronic Neuron

IEEE JOURNAL OF QUANTUM ELECTRONICS

De Marinis, Lorenzo; Catania, Alessandro; Castoldi, Piero; Contestabile, Giampiero; Bruschi, Paolo; Piotto, Massimo; Andriolli, Nicola

被引用次数: 12

2022

Low-Phase-Noise CMOS Relaxation Oscillators for On-Chip Timing of IoT Sensing Platforms

ELECTRONICS

Gagliardi, Francesco; Manfredini, Giuseppe; Ria, Andrea; Piotto, Massimo; Bruschi, Paolo

被引用次数: 6

2022

Design of a Capacitance-to-Digital Converter Based on Iterative Delay-Chain Discharge in 180 nm CMOS Technology

SENSORS

Cicalini, Mattia; Piotto, Massimo; Bruschi, Paolo; Dei, Michele

被引用次数: 3

2022

Dual-Band Ge-on-Si Photodetector Array With Custom, Integrated Readout Electronics

IEEE SENSORS JOURNAL

De Iacovo, Andrea; Mitri, Federica; Ballabio, Andrea; Frigerio, Jacopo; Isella, Giovanni; Ria, Andrea; Cicalini, Mattia; Bruschi, Paolo; Colace, Lorenzo

被引用次数: 2

2022

A 3-decade-frequency-range Sinewave Synthesizer with Analog Piecewise-linear Interpolation

PRIME 2022: 17TH INTERNATIONAL CONFERENCE ON PHD RESEARCH IN MICROELECTRONICS AND ELECTRONICS

Gagliardi, F.; Ria, A.; Manfredini, G.; Bruschi, P.; Piotto, M.

被引用次数: 0

2022

A Triode-Compensated CMOS Bandgap Core for Sub-250 mV Supply Voltages

PRIME 2022: 17TH INTERNATIONAL CONFERENCE ON PHD RESEARCH IN MICROELECTRONICS AND ELECTRONICS

Gagliardi, F.; Ria, A.; Manfredini, G.; Piotto, M.; Bruschi, P.

被引用次数: 0

2022

An Analog Driver compliant with Supply Voltages exceeding the Ratings of Standard CMOS Processes

2022 29TH IEEE INTERNATIONAL CONFERENCE ON ELECTRONICS, CIRCUITS AND SYSTEMS (IEEE ICECS 2022)

Gagliardi, Francesco; Cicalini, Mattia; Ria, Andrea; Piotto, Massimo; Bruschi, Paolo

被引用次数: 1

2022

Management and Sustainable Exploitation of Marine Environments through Smart Monitoring and Automation

JOURNAL OF MARINE SCIENCE AND ENGINEERING

Glaviano, Francesca; Esposito, Roberta; Cosmo, Anna Di; Esposito, Francesco; Gerevini, Luca; Ria, Andrea; Molinara, Mario; Bruschi, Paolo; Costantini, Maria; Zupo, Valerio

被引用次数: 27

2022

An ASIC-Based Miniaturized System for Online Multi-Measurand Monitoring of Lithium-Ion Batteries

BATTERIES-BASEL

Manfredini, Giuseppe; Ria, Andrea; Bruschi, Paolo; Gerevini, Luca; Vitelli, Michele; Molinara, Mario; Piotto, Massimo

Modeling and optimization of directive acoustical particle velocity sensors for ultrasonic applications

SENSORS AND ACTUATORS A-PHYSICAL

Benvenuti, L.; Catania, A.; Bruschi, P.; Piotta, M.

被引用次数: 1

2021

Design Strategies and Architectures for Ultra-Low-Voltage Delta-Sigma ADCs

ELECTRONICS

Benvenuti, Lorenzo; Catania, Alessandro; Manfredini, Giuseppe; Ria, Andrea; Piotta, Massimo; Bruschi, Paolo

被引用次数: 3

2021

Energy Efficiency in Slew-Rate Enhanced Single-Stage OTAs for Switched-Capacitor Applications

JOURNAL OF LOW POWER ELECTRONICS AND APPLICATIONS

Catania, Alessandro; Cicalini, Mattia; Piotta, Massimo; Bruschi, Paolo; Dei, Michele

被引用次数: 3

2021

A False Positive Reduction System For Continuous Water Quality Monitoring

2021 IEEE INTERNATIONAL CONFERENCE ON SMART COMPUTING (SMARTCOMP 2021)

Bria, A.; Ferrigno, L.; Gerevini, L.; Marrocco, C.; Molinara, M.; Bruschi, P.; Cicalini, M.; Manfredini, G.; Ria, A.; Cerro, G.; Simmarano, R.; Teolis, G.; Vitelli, M.

被引用次数: 0

2021

A Codesigned Photonic Electronic MAC Neuron with ADC-Embedded Nonlinearity

2021 CONFERENCE ON LASERS AND ELECTRO-OPTICS (CLEO)

De Marinis, L.; Catania, A.; Castoldi, P.; Bruschi, P.; Piotta, M.; Andriolli, N.

被引用次数: 0

2021

A Low-Power CMOS Bandgap Voltage Reference for Supply Voltages Down to 0.5 V

ELECTRONICS

Ria, Andrea; Catania, Alessandro; Bruschi, Paolo; Piotta, Massimo

被引用次数: 7

2021

Experimental Evaluation of a 3D-Printed Fluidic System for a Directional Anemometer

SENSORS

Ria, Andrea; Catania, Alessandro; Bruschi, Paolo; Piotta, Massimo

被引用次数: 3

2020

Ultra-Low-Voltage Inverter-Based Amplifier with Novel Common-Mode Stabilization Loop

ELECTRONICS

Manfredini, Giuseppe; Catania, Alessandro; Benvenuti, Lorenzo; Cicalini, Mattia; Piotta, Massimo; Bruschi, Paolo

被引用次数: 22

2020

Performance Analysis and Design Optimization of Parallel-Type Slew-Rate Enhancers for Switched-Capacitor Applications

ELECTRONICS

Catania, Alessandro; Cicalini, Mattia; Dei, Michele; Piotto, Massimo; Bruschi, Paolo

被引用次数: 2

2020

Thermal Noise-Boosting Effects in Hot-Wire-Based Micro Sensors

JOURNAL OF SENSORS

Piotto, Massimo; Catania, Alessandro; Nannini, Andrea; Bruschi, Paolo

被引用次数: 3

2020

A 2 nW 0.25 V 140 dB-FOM Inverter-Based First Order $\Delta\Sigma$ Modulator

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Catania, Alessandro; Benvenuti, Lorenzo; Ria, Andrea; Manfredini, Giuseppe; Piotto, Massimo; Bruschi, Paolo

被引用次数: 10

2020

A Two-Stage Switched-Capacitor Integrator for High Gain Inverter-Like Architectures

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Bruschi, P.; Catania, A.; Del Cesta, S.; Piotto, M.

被引用次数: 10

2020

CMOS Interfaces for Internet-of-Wearables Electrochemical Sensors: Trends and Challenges

ELECTRONICS

Dei, Michele; Aymerich, Joan; Piotto, Massimo; Bruschi, Paolo; Javier del Campo, Francisco; Serra-Graells, Francesc

被引用次数: 17

2019

A Compact Low-Offset Instrumentation Amplifier with Wide Input and Output Ranges

SENSORS

Piotto, Massimo; Del Cesta, Simone; Argenio, Giovanni; Simmarano, Roberto; Bruschi, Paolo

被引用次数: 0

2018

Integrated Thermal Flow Sensors with Programmable Power-Sensitivity Trade-Off

SENSORS

Piotto, Massimo; Dell'Agnello, Filippo; Del Cesta, Simone; Bruschi, Paolo

被引用次数: 0

2018

A compact current-mode instrumentation amplifier for general-purpose sensor interfaces

AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS

Del Cesta, Simone; Ria, Andrea; Piotto, Massimo; Simmarano, Roberto; Bruschi, Paolo

被引用次数: 4

2018

Determination of the Wind Speed and Direction by Means of Fluidic-Domain Signal Processing

A CMOS compatible micro-Pirani vacuum sensor based on mutual heat transfer with 5-decade operating range and 0.3 Pa detection limit

SENSORS AND ACTUATORS A-PHYSICAL

Piotto, Massimo; Del Cesta, Simone; Bruschi, Paolo

被引用次数: 12

2017

A Chopper Instrumentation Amplifier With Input Resistance Boosting by Means of Synchronous Dynamic Element Matching

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Butti, Federico; Piotto, Massimo; Bruschi, Paolo

被引用次数: 44

2017

Precise Measurement of Gas Volumes by Means of Low-Offset MEMS Flow Sensors with $\mu\text{L}/\text{min}$ Resolution

SENSORS

Piotto, Massimo; Del Cesta, Simone; Bruschi, Paolo

被引用次数: 5

2017

Integrated smart gas flow sensor with 2.6 mW total power consumption and 80 dB dynamic range

MICROELECTRONIC ENGINEERING

Piotto, Massimo; Del Cesta, Francesco; Bruschi, Paolo

被引用次数: 10

2016

Characterization and modeling of CMOS-compatible acoustical particle velocity sensors for applications requiring low supply voltages

SENSORS AND ACTUATORS A-PHYSICAL

Piotto, M.; Butti, F.; Zanetti, E.; Di Pancrazio, A.; Iannaccone, G.; Bruschi, P.

被引用次数: 8

2015

An Integrated Thermal Flow Sensor for Liquids Based on a Novel Technique for Electrical Insulation

SENSORS

Piotto, Massimo; Di Pancrazio, Alessia; Intaschi, Luca; Bruschi, Paolo

被引用次数: 0

2015

A Novel Compact Instrumentation Amplifier for Optimal Interfacing of Thermoelectric Sensors

SENSORS

Piotto, M.; Butti, F.; Del Cesta, F.; Longhitano, A. N.; Bruschi, P.

被引用次数: 0

2015

A Low-Cost Sensing System for Cooperative Air Quality Monitoring in Urban Areas

SENSORS

Brienza, Simone; Galli, Andrea; Anastasi, Giuseppe; Bruschi, Paolo

被引用次数: 57

2015

Automatic compensation of pressure effects on smart flow sensors in the analog and digital domain

SENSORS AND ACTUATORS A-PHYSICAL

Piotto, M.; Longhitano, A. N.; Del Cesta, F.; Bruschi, P.

被引用次数: 7

2014

Seebeck Coefficient of Nanowires Interconnected into Large Area Networks

NANO LETTERS

Pennelli, Giovanni; Totaro, Massimo; Piotto, Massimo; Bruschi, Paolo

被引用次数: 30

2013

A Low-Power Interface for Capacitive Sensors With PWM Output and Intrinsic Low Pass Characteristic

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Nizza, Nicolò; Dei, Michele; Butti, Federico; Bruschi, Paolo

被引用次数: 49

2013

A compact instrumentation amplifier for MEMS thermal sensor interfacing

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Butti, Federico; Bruschi, Paolo; Dei, Michele; Piotto, Massimo

被引用次数: 12

2012

Design Issues for Low Power Integrated Thermal Flow Sensors with Ultra-Wide Dynamic Range and Low Insertion Loss

MICROMACHINES

Bruschi, Paolo; Piotto, Massimo

被引用次数: 21

2012

Smart Flow Sensor With On-Chip CMOS Interface Performing Offset and Pressure Effect Compensation

IEEE SENSORS JOURNAL

Piotto, Massimo; Dei, Michele; Butti, Federico; Pennelli, Giovanni; Bruschi, Paolo

被引用次数: 23

2012

Top down fabricated silicon nanowire networks for thermoelectric applications

MICROELECTRONIC ENGINEERING

Totaro, M.; Bruschi, P.; Pennelli, G.

被引用次数: 19

2012

Fabrication and characterization of a directional anemometer based on a single chip MEMS flow sensor

MICROELECTRONIC ENGINEERING

Piotto, M.; Pennelli, G.; Bruschi, P.

被引用次数: 26

2011

Surface roughness and electron backscattering in high aspect ratio silicon nanowires

MICROELECTRONIC ENGINEERING

Pennelli, G.; Totaro, M.; Bruschi, P.

被引用次数: 1

2011

An Offset Compensation Method With Low Residual Drift for Integrated Thermal Flow Sensors

IEEE SENSORS JOURNAL

Bruschi, Paolo; Dei, Michele; Piotto, Massimo

被引用次数: 28

2011

A Method to Compensate the Pressure Sensitivity of Integrated Thermal Flow Sensors

IEEE SENSORS JOURNAL

Bruschi, Paolo; Dei, Michele; Piotto, Massimo

被引用次数: 8

2010

Validation of the Compatibility Between a Porous Silicon-Based Gas Sensor Technology and Standard Microelectronic Process

IEEE SENSORS JOURNAL

Barillaro, G.; Bruschi, P.; Lazzerini, G. M.; Strambini, L. M.

被引用次数: 21

2010

A single chip, double channel thermal flow meter

MICROSYSTEM TECHNOLOGIES-MICRO-AND NANOSYSTEMS-INFORMATION STORAGE AND PROCESSING SYSTEMS

Bruschi, P.; Dei, M.; Piotto, M.

被引用次数: 39

2009

A voltage controlled CMOS current divider with linear characteristic

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Dei, Michele; Nizza, Nicolo; Piotto, Massimo; Bruschi, Paolo

被引用次数: 1

2009

Postprocessing, readout and packaging methods for integrated gas flow sensors

MICROELECTRONICS JOURNAL

Bruschi, P.; Piotto, M.; Bacci, N.

被引用次数: 12

2009

A Low-Power 2-D Wind Sensor Based on Integrated Flow Meters

IEEE SENSORS JOURNAL

Bruschi, Paolo; Dei, Michele; Piotto, Massimo

被引用次数: 57

2009

A compact package for integrated silicon thermal gas flow meters

MICROSYSTEM TECHNOLOGIES-MICRO-AND NANOSYSTEMS-INFORMATION STORAGE AND PROCESSING SYSTEMS

Bruschi, P.; Nurra, V.; Piotta, M.

被引用次数: 20

2008

Design and analysis of integrated flow sensors by means of a two-dimensional finite element model

SENSORS AND ACTUATORS A-PHYSICAL

Bruschi, P.; Ciomei, A.; Piotta, M.

被引用次数: 11

2008

A current-mode, dual slope, integrated capacitance-to-pulse duration converter

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bruschi, Paolo; Nizza, Nicolo; Piotta, Massimo

被引用次数: 50

2007

Measurement and modeling of pulsed mode flow meter for liquids based on a single chip probe

SENSORS AND ACTUATORS A-PHYSICAL

Bruschi, P.; Nizza, N.; Piotta, M.

被引用次数: 5

2006

Effects of gas type on the sensitivity and transition pressure of integrated thermal flow sensors

SENSORS AND ACTUATORS A-PHYSICAL

Bruschi, P.; Piotta, M.; Barillaro, G.

被引用次数: 20

2006

CMOS transconductors with nearly Constant input ranges over wide tuning intervals

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Bruschi, Paolo; Sebastiano, Fabio; Nizza, Nicolo

被引用次数: 8

2006

A double heater integrated gas flow sensor with thermal feedback

SENSORS AND ACTUATORS A-PHYSICAL

Bruschi, P.; Diligenti, A.; Navarrini, D.; Piotta, M

被引用次数: 53

2005

A method for cross-sensitivity and pull-in voltage measurement of MEMS two-axis accelerometers

SENSORS AND ACTUATORS A-PHYSICAL

Bruschi, P.; Nannini, A.; Paci, D.; Pieri, F

被引用次数: 17

2005

A precise capacitance-to-pulse width converter for integrated sensors

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Bruschi, P.; Navarrini, D.; Barillaro, G.; Gola, A

被引用次数: 2

2005

A flow sensor for liquids based on a single temperature sensor operated in pulsed mode

SENSORS AND ACTUATORS A-PHYSICAL

Bruschi, P; Navarrini, D; Piotto, M

被引用次数: 6

2004

Sensitivity improvement of integrated thermal anemometers obtained by jet flow impingement

SENSORS AND ACTUATORS A-PHYSICAL

Bruschi, P; Navarrini, D; Piotto, M; Raffa, G

被引用次数: 4

2004

Electrical measurements of the quality factor of microresonators and its dependence on the pressure

SENSORS AND ACTUATORS A-PHYSICAL

Bruschi, P; Nannini, A; Pieri, F

被引用次数: 18

2004

A precise capacitance-to-pulse width converter for integrated sensors

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Bruschi, P; Navarrini, D; Barillaro, G; Gola, A

被引用次数: 3

2004

A high current drive CMOS output stage with a tunable quiescent current limiting circuit

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Bruschi, P; Navarrini, D; Piotto, M

被引用次数: 7

2003

Two-dimensional macroscopical simulations of porous silicon growth

COMPUTATIONAL MATERIALS SCIENCE

Barillaro, G; Bruschi, P; Pieri, F

被引用次数: 6

2002

Micromachined gas flow regulator for ion propulsion systems

IEEE TRANSACTIONS ON AEROSPACE AND ELECTRONIC SYSTEMS

Bruschi, P; Diligenti, A; Piotto, M

被引用次数: 12

2002

Flicker noise in heterocyclic conducting polymer thin film resistors

FLUCTUATION AND NOISE LETTERS

Bruschi, P; Nannini, A; Navarrini, D; Piotto, M

被引用次数: 2

2002

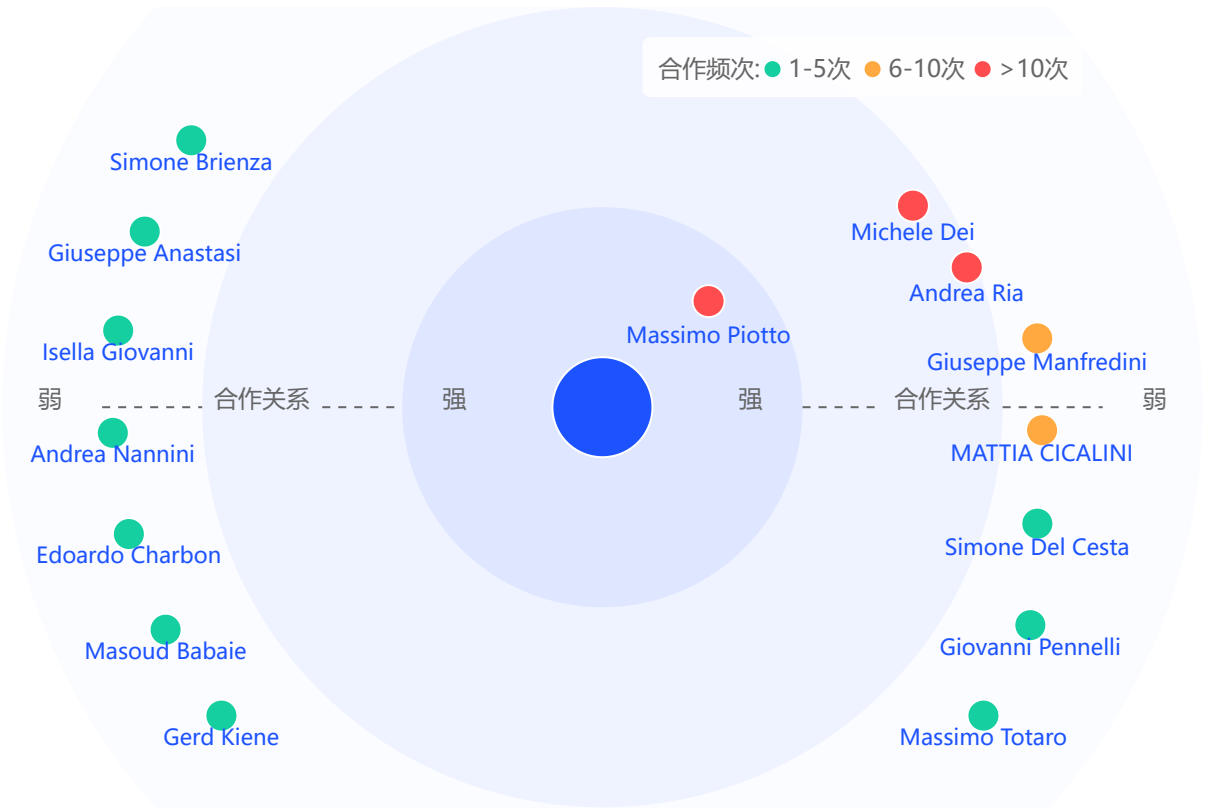
Micromachined silicon suspended wires with submicrometric dimensions

MICROELECTRONIC ENGINEERING

Bruschi, P; Diligenti, A; Piotto, M

被引用次数：7	2001
Three-dimensional Monte Carlo Simulations of electromigration in polycrystalline thin films	
COMPUTATIONAL MATERIALS SCIENCE	
Bruschi, P; Nannini, A; Piotto, M	
被引用次数：31	2000
Technology of integrable free-standing yttria-stabilized zirconia membranes	
THIN SOLID FILMS	
Bruschi, P; Diligenti, A; Nannini, A; Piotto, M	
被引用次数：11	1999
Simulation of the deposition and aging of thin island films	
COMPUTATIONAL MATERIALS SCIENCE	
Bruschi, P; Nannini, A	
被引用次数：0	1998

学术合作



Massimo Piotto	合作次数：35
比萨大学	
Michele Dei	合作次数：13

比萨大学

Andrea Ria

合作次数: 11

比萨大学

Giuseppe Manfredini

合作次数: 6

比萨大学

MATTIA CICALINI

合作次数: 6

比萨大学

还有 10 个合作者